THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

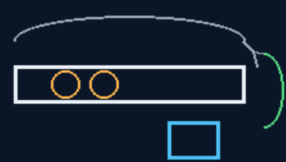
THE GEMINI PROTOCOLS

PHASE 250

THE MAGNETIC SHAKER, UPSIZED
GAUSS-PRIMED MOMENTUM ENGINE

pin-primed, photo-gated, self-reloading

air for the punch, vacuum for the journey



no ball ever touches a neo — locked 4 August 2026

THE SEESAW LAW · LENZ RIDES THE CARRIAGE

RATHER THE MINI LOSS

Momentum engine — v1.0 in force

GP-IM-250

Revision v1.0 — 24 August 2026

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VETTED BATCH 35 — PHYSICS PASS

PRIMARY AUTHOR & INVENTOR: ARCHER QUINN — AEROSPACE DESIGN ENGINEER, NPHD

CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 250

PHASE 250: THE MAGNETIC SHAKER WORK TORCH, UPSIZED — GAUSS-PRIMED ZERO-G MOMENTUM ENGINE — STATUS: CURRENT — PHYSICS PASS / EFFICIENCY, MAGNETIC GEOMETRY, SWITCHING TIMING, IMPACT ENERGY AND MECHANICAL FATIGUE TESTING (VET BATCH 35). The shaker torch grows up into a momentum engine. Pin-primed, photo-gated, self-reloading: the gauss gun fires two balls, the loop brings them home on the carriage, the reload runs both ways, the gated ring holds the door. The details were seated verbatim in order of the catch: the return slope ('slope thing yes but i have counter momentum no much but in zero g enough to roll it'), the catch spring ('spring on the carriage?' — answered yes: kills the clack tax), the seat we missed (the side tapper — tiny pin or lever for a side flick, no gravity carriage), the tube geometry (wall bearings same as its wheels), the scale (gun is 3 x 50 kg-power neos — crush-your-hand power — balls, carriage and coils scaled). The vacuum split resolution: air for the punch, vacuum for the journey, the ring holds the door. The physics ledger: the amplifier is a gearbox, the seesaw law prices the rebound, Lenz rides the carriage. THE NEO PROTECTION LAW, locked by the author 4 August 2026: no ball ever touches a neo — steel between, spring between, sacrificial faces everywhere; rather the mini loss than deep space unreplaceable broken neos.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-250, v1.0 — 24 August 2026. The two hundred and fifty-first manual of the program — 251 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the engine law as seated (contents line, the crayon preface, the machine verbatim, the details in order of the catch, the vacuum split, the physics note, the neo protection law, the bench items) — §2 the physics, graded — §3 the system in the loop — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status and provenance.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 250: **The Magnetic Shaker Work Torch, Upsized — Gauss-Primed Zero-G Momentum Engine: Pin-Primed, Photo-Gated, Self-Reloading; the Neo Protection Law Locked — No Ball Ever Touches a Neo).**'

THE CORE OBJECTIVE: none seated as a standalone line — the contents line above carries the register wording.

THE CRAYON PREFACE — SEATED AS ORDERED (THE HONEST FRAME) (VERBATIM)

The author's frame, verbatim: "you see im just rebuilding the shaker torch using magnets instead of my hands to get something too big to shake but that means more out, in fact i would call it a magnetic shaker and preface it as the shaker torch upsized, so its not some perpetual motion first instinct for the crayon readers"

The Magnetic Shaker — the shaker torch, upsized. In the forever torch (241), your arm shakes the magnet through the coil and your arm pays the bill. Here the arm is too small for the job, so a pin and a gated coil do the shaking — and the ship's grid pays the bill exactly like your arm did, only bigger. The magnet stages between them are the gearbox: they take a small electric shove and deliver one big shake. More shake out, because more shake in. Nothing else about the physics changed.

THE MACHINE (THE AUTHOR'S WORDS, SEATED VERBATIM) (VERBATIM)

THE GUN: "gausin gun, 2 balls 1 hits from behind makes the one in front fire, the more magnets between the more it fires, a free amplifier"

THE LOOP & THE CARRIAGE: "now front ball fires around a tube loop coiled of course in zero g, it hits a small carriage in from that has a ball magnet, back of the carriage is long enough to stop the ball sticking, track has a slight slope ball runs back to the gun after inertia transfer."

THE RELOAD: "the firing ball. sits at the back of the gun magnets, a push pin fire up pushing it up a tube sitting ball clearance above it, , zero g it fires around a vertical loop into the back of the gun again."

BOTH WAYS: "reload is covered both ways, the charge engine carriages goes around its loop back to where it started right beside its starting point, the end stop is sloped inwards, the carriage wheels are cup balls non magnetic, so it rolls in any direction."

THE GATED RING: "pin fires ball up over loop crashes into the back of the magnet set firing the front ball. the magnet set is all neos except the one at the strike point that turns on photo electric right before the shrike and off by photo electric from the front ball moving. the power in zero g is mostly the length of the track as to output."

THE DETAILS (SEATED VERBATIM, IN ORDER OF THE CATCH) (VERBATIM)

- THE RETURN: "slope thing yes but i have counter momentum no much but in zero g enough to roll it back" — affirmed: in zero-g there is no hill; the rebound whisper is a full fare, friction is the only toll.
- THE CATCH SPRING: "spring on the carriage?" — answered yes: kills the clack tax, softens the impulse, rates the transfer; Belleville stack for short travel, steel not elastomer; the 175 magnetic spring is the contactless deluxe version. The spring changes the tax, not the split.
- THE SEAT: "we missed the side tapper, tiny pin or lever for a side flick, no gravity carriage wont roll" — the last gravity part caught. Options seated: detent latch (passive, seats itself), weak magnetic seat (the gun's own trick), or the tapper (active, in the pin family).
- THE TUBE GEOMETRY: "ok carriage needs wall bearings same as its wheels, the rear bumper can be narrower than the main carriage slightly, limiting wall contact over length, front rocket conical, no front edges catching wall." — only the bearings touch; wall cup-balls run light preload in zero-g.
- THE SCALE: "gun is 3 x 50kg power neos (crush you hand power) balls, carriage and coils scaled to match probably 50-60mm if i recall size, so the carriage must weigh equal total to the bullet" — and the tuning question: "what is the percentage weight increase for the carriage 5%"

THE VACUUM SPLIT (SEATED VERBATIM + THE AUDIT RESOLUTION) (VERBATIM)

"ok so vacuum adds features to some kills others, just spitballing, across the back off the gun put the rear section, because they dont need to be in the same box, in its own vacuum, gain or loss at the strike?"

"so strike plate of the electromagnet is the vac box physical wall, the gun stays open air. worth the effort, or leave open air"

- RESOLUTION (KIMI AUDIT): air for the punch, vacuum for the journey, the ring holds the door. The strike stays in air — cold welding never gets a vote; the flight sections go in vacuum — the tube-piston drag tax deleted. The ring electromagnet's plate is the pressure boundary; the bore is the only leak. Bench version: open air, everything touchable. Ship version at long track and high cycle rate: pump the box. Riders: ceramic or dry-film casters in vacuum, photogates sighted through a viewport or vacuum-rated feedthroughs, Paschen never gets a vote (every volt stays in air).

PHYSICS NOTE — THE AUDIT (THE LEDGER, AFFIRMED AND BOUND) (VERBATIM)

- **The amplifier is a gearbox.** Gauss staging amplifies velocity per trigger, not energy per cycle. The gain comes from the well-depth difference between seats; restoring the deep-captured ball to the shallow seat costs exactly the gain. The reseat pays the bill — the file's own ledger, line 4. The photo-gated electromagnet does not repeal it; it moves part of the bill to the wall, which is the honest way to pay.

- **The turn point is real** (the author's bench memory, affirmed): each stage adds a fixed energy, so velocity grows as the square root of stage count; impact losses climb with speed; and past a strike threshold the neos shatter. The practical plateau sits below the theoretical one. Deeper wells raise the curve; the active coil feeds it; gun length only counts where the coil lives; cycle rate multiplies average power.
- **The seesaw law.** Rebound fraction = (M-m)/(M+m); harvest = 4mM/(m+M)^2. Equal mass = *[WORDING KILLED 15 AUGUST 2026 — author's order 'kill anything that says absolute or 100 percent'; original: '100%']* harvest and zero return. At the author's 5% setting: 2.4% rebound, 99.9% harvest — the greedy end, and the correct place to start; carriage weight is the tuning knob (10% -> 4.8%/99.8%; 20% -> 9%/99%; 50% -> 20%/96%; double -> 33%/89%).
- **Lenz rides the carriage.** The charge engine's generation IS its braking: it comes home with strike minus friction minus what it generated. "Takes nothing to spin" applies only unloaded, and the charge engine is loaded by definition.
- **Honesty for the crayon readers.** Throughput, not efficiency: the grid's bill scales with the coil exactly like the dinner bill scaled with the arm. The machine works while you sleep — that is its whole claim, and it needs no other.
- **Scale numbers (bench estimates).** Three 50 kg-pull neos make a ~1,500 N stack; wells ~10-15 J per stage; realistic 10-25 J to the projectile after the plateau; a 55 mm steel ball is ~0.7 kg leaving at 6-8 m/s. A hammer in a tube — non-magnetic tools inside a meter, spacers between neos, face shield for assembly, nothing ferrous loose on the bench.

THE NEO PROTECTION LAW — LOCKED BY THE AUTHOR, 4 AUGUST 2026 (VERBATIM)

The reseat clack, caught by the author: "actually there is a neo impact, slight but when the ball returns and hits the front neo field it will get pulled back in hard, but nothing like the strike end at high speed"

The coating truth, the author's: "yes you are right about the neos, the flaw is in the coating, as size increases the coating only increase slightly, that is the tipping point i think, 25kilo neos likely have same coating thickness {shell} as a 50kilo"

THE LOCK, VERBATIM: "lets lock it in at that, rather the mini loss than deep space unreplaceable broken neos"

- **THE LAW: no ball ever touches a neo.** Steel between, spring between, sacrificial faces everywhere contact lives; the electromagnet's strike plate is the anvil; the front seat gets a steel face. The coating is corrosion armor, not impact armor — armor is architecture, not plating.

- **The mini loss is accepted as law.** The eddy collar covers the reseat: velocity-selective (brutal on the whisper, brushes the bullet at ~2-3%), one-way by geometry on the return leg. A few percent is cheap insurance; a shattered magnet in deep space is forever.

BENCH ITEMS (THE DATA LOG DOCTRINE — BENCH IT, LOG IT, THE CURVES ARGUE) (VERBATIM)

- **Impact-to-shatter:** "lets play catch, start at 25" — start at 25 kg neos, ramp the strike energy, log first-crack and full-shatter per rating, in containment (the shrapnel is sharp and still magnetic). Log the assembly, not just the magnet — strike plate and spring change the peak stress the substrate ever sees.
- **Return-loop friction log:** how much whisper the loop eats — sets the mass-ratio ladder setting.
- **Collar sizing:** v-in vs v-out at reseat and firing speeds; target — whisper exits near zero, bullet exits at 97+ percent.
- **The cycle ledger:** pin energy in, coil energy out, per cycle, on paper. The standing offer for every machine in this file.

ORIGIN OF THOUGHT — PHASE 250 (VERBATIM)

One sitting, waiting on the mailman: the gauss gun, the loops, the carriage, the spring, the side tapper, the bearings, the masses, the coating truth, the collar, the lock. The machine confessed its gravity parts one by one and armored every contact. Lineage: 241 (the shaker torch — Amendment Zero of this machine), 79 (the powder), 175 (the magnetic spring), 249 (the extraction bake-off — the charge-engine carriage is its dumb-rotor cousin). The upload, when it lands, amends this phase.

INSTRUCTION STATUS (10 August 2026, sitting 147): PARTIAL — the author's voice seated in this body is the surviving fragment of the instruction; the full live order and its thread were not preserved (the 200–260 band was rebuilt by the audit without the chat record). The author counts this phase among the 80 lost instructions. The seated voice is the instruction of record; the audit prose around it is scaffolding.

- **VET FLAG — BATCH 4 (10 August 2026): TERMINOLOGY.** Physics passes — the grid pays the energy bill exactly as the arm pays in the hand-shaken torch. But "free amplifier" is dangerous wording for machine readers; the vet's machine-safe terms: magnetic force amplifier / mechanical stroke amplifier. Awaiting the author's notation pass. *[RESOLVED 11 August 2026 — AUTHOR'S NOTATION PASS SEATED BELOW]*
- **AMENDMENT — AUTHOR'S NOTATION PASS, 11 August 2026 (flag fixed on the author's order: "fix the flags").** The vet's machine-safe terms are ADOPTED: this machine is a MAGNETIC FORCE AMPLIFIER / MECHANICAL STROKE AMPLIFIER. The word 'free' is struck from the specification vocabulary for this machine — the grid pays the energy bill exactly as the arm pays in the hand-

shaken torch; force is amplified, energy is not created. The historical wording stays preserved verbatim in the transcript region.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE AMPLIFIER IS A GEARBOX, AFFIRMED AND BOUND: gauss staging amplifies velocity per trigger, not energy per cycle — the grid pays the energy bill, the machine pays it out as speed. The physics note seats the ledger in crayon: throughput, not efficiency; the grid's bill scales with the work, and the file says so for the crayon readers.

THE TURN POINT IS REAL — THE AUTHOR'S BENCH MEMORY, AFFIRMED: each stage adds a fixed energy; beyond the turn point the gains flatten against the losses. The bench memory and the physics note agree; the design lives under the turn point.

THE SEESAW LAW, SEATED: rebound fraction = (M-m)/(M+m); harvest = 4mM/(m+M)^2. Equal mass is the optimum the family keeps rediscovering; the mass-ratio ladder is set by the return-loop friction log, not by hope.

LENZ RIDES THE CARRIAGE: the charge engine's generation IS its braking — it comes home with less than it left with, every cycle, and the difference is the harvest. The vacuum split resolution seats the operating mode: air for the punch, vacuum for the journey, the ring holds the door.

THE MACHINE, VERBATIM IN ORDER OF THE CATCH: the return slope with its counter-momentum; the catch spring ('spring on the carriage?' — yes: kills the clack tax, softens the impact); the side tapper ('we missed the side tapper, tiny pin or lever for a side flick, no gravity carriage'); the tube geometry (wall bearings same as its wheels, the rear bumper); the scale (3 x 50 kg-power neos, crush-your-hand power, balls carriage and coils scaled). His words, seated in order.

THE NEO PROTECTION LAW — LOCKED BY THE AUTHOR, 4 AUGUST 2026: no ball ever touches a neo. Steel between, spring between, sacrificial faces everywhere. The mini loss is accepted as law — the eddy collar covers the reseat, velocity-selective. The lock, verbatim: 'lets lock it in at that, rather the mini loss than deep space unreplaceable broken neos'.

THE SCALE NUMBERS (BENCH ESTIMATES, SEATED AS ESTIMATES): three 50 kg-pull neos make about a 1,500 N stack; wells about 10-15 mm; the impact-to-shatter bench starts at 25 kg neos and ramps the strike energy — 'lets play catch, start at 25', verbatim.

§3 — THE SYSTEM IN THE LOOP

- **The gun:** gauss-primed, pin-primed, photo-gated — two balls, self-reloading both ways.
- **The loop:** the carriage brings the balls home — wall bearings same as its wheels, the rear bumper.

- **The catches:** the catch spring kills the clack tax; the side tapper gives the flick a no-gravity carriage needs.
- **The split:** air for the punch, vacuum for the journey — the gated ring holds the door.
- **The scale:** 3 x 50 kg-power neos, crush-your-hand power — about a 1,500 N stack; start the bench at 25 kg.
- **The law:** no ball ever touches a neo — steel between, spring between, sacrificial faces everywhere; locked 4 August 2026.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- The shaker lineage closes: 241's hand torch and the upsized engine are one machine at two scales — the same coil-and-magnet heart, one in the fist, one in the floor.
- The neo protection law is fleet law now: locked 4 August 2026, quoted verbatim, and binding on every design in the band — rather the mini loss than deep space unreplaceable broken neos.
- The data-log doctrine is the phase's own: bench it, log it, the curves argue — the cycle ledger (pin energy in, coil energy out, per cycle, on paper) is the standing offer for any challenger.
- The vet flag record is carried honestly: batch 4 flagged terminology — physics passes; the 11 August 2026 notation pass fixed it on the author's order, and both the flag and the fix are seated.

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT), VET BATCH 35 (17 August 2026 — phases 246-250 plus the 247 rebuttal exchange, cross-checked in sitting 404), SEATED. The grade, verbatim from the batch: '250 → physics PASS / efficiency, magnetic geometry, switching timing, impact energy and mechanical fatigue testing.' The remaining tests named by the verdict itself: efficiency, magnetic geometry, switching timing, impact energy and mechanical fatigue testing.

§6 — BUILD SEQUENCE

- 1. Build the gun: pin-primed, photo-gated staging — switching timing benched before scale-up.
- 2. Close the loop: carriage with wall bearings, the return slope, the catch spring, the rear bumper.
- 3. Fit the flick: the side tapper — tiny pin or lever; the no-gravity carriage's missing seat.
- 4. Armour the neos: steel between, spring between, sacrificial faces — the protection law applied to every contact.

- 5. Run the ledger: start at 25 kg neos, ramp the strike energy, log impact-to-shatter, return-loop friction, collar sizing, the cycle ledger — publish the curves.

§7 — CROSS-PHASE (INLINED IN FULL)

- **Phase 241 (the Forever Light Stick):** the parent — the hand shaker's coil-and-magnet heart, upsized into the floor machine; the family at two scales.
- **Phase 249 (the orbital engine):** the cousin doctrine — stored labour honestly named; the gearbox framing and the crank framing are the same honesty at two speeds.
- **Phase 251 (the tow truck slingshot):** the momentum family — the seesaw law prices both the carriage rebound and the bolo catch; play catch is the family law.

§8 — DO-NOT-BUILD

- Do NOT let any ball touch a neo — the law is locked: steel between, spring between, sacrificial faces; rather the mini loss.
- Do NOT sell efficiency — throughput, not efficiency; the grid pays the bill and the crayon preface says so.
- Do NOT skip the side tapper — the no-gravity carriage needs the flick; we missed it once, the file says so.
- Do NOT bench past the protection — start at 25 kg, ramp; impact-to-shatter is measured, never assumed.
- Do NOT run the punch in vacuum — air for the punch, vacuum for the journey; the ring holds the door, the split is law.

§9 — OWED

OWED: the vet's five, whole — efficiency mapping; magnetic geometry optimisation; switching timing; impact energy (the shatter bench from 25 kg up); mechanical fatigue testing. Plus the cycle ledger kept per doctrine: pin energy in, coil energy out, per cycle, on paper, published.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-250, v1.0 — CURRENT — PHYSICS PASS / EFFICIENCY, MAGNETIC GEOMETRY, SWITCHING TIMING, IMPACT ENERGY AND MECHANICAL FATIGUE TESTING (VET BATCH 35), seated law, master v2.1 (numerical print order). The two hundred and fifty-first manual of the program — 251 of 290. Built 24 August 2026, eleventh of the 20-manual 'ok next 20' shift (the author's order, 24 August

2026, seated in sitting 622). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590, 597, 607, 618 and 622): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the 200' / 'ok i will do last nights now you do the next 20' / 'ok ill do these 20, you do 20 more' / 'ok next 20'. The phase's own chain, all seated in the master: the contents line, the combined masthead line and the bodies (as seated); the live instruction of 10 August 2026 (seated per sitting 147, PARTIAL — the surviving fragment) — the machine and the details seated verbatim in order of the catch; the vacuum split resolution (air for the punch, vacuum for the journey, the ring holds the door); the neo protection law, locked by the author 4 August 2026, verbatim: 'lets lock it in at that, rather the mini loss than deep space unreplaceable broken neos'; the rename of 12 August 2026; the batch 4 terminology flag and the 11 August 2026 notation pass; the vet batch 35 grade (physics PASS / efficiency, magnetic geometry, switching timing, impact energy and mechanical fatigue testing) in §5.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the builders'). Next update triggers: the impact-to-shatter curve and the cycle ledger, or his word.

END OF MANUAL — PHASE 250